Local Spectral Thermodynamics and the Einstein Equation
as a Spectral Equilibrium Condition

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Version beta2

Abstract

We develop a local thermodynamic interpretation of the projective spectral entropy functional $S_\Pi[g] = \frac{1}{2}\log\det' A_g$, extending the covariant framework established in a companion paper. The diagonal heat kernel $K(x,x;t)$ defines a local spectral energy density $u(x;t) = -\partial_t \log K(x,x;t)$, whose Seeley–DeWitt expansion yields a local spectral multiplier field $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R)$. Here $\beta_0^{-1} = 2/t_*$ is a universal UV contribution fixed by the spectral admissibility scale t_*, while the curvature correction is geometric and covariant.

We then formulate a local spectral first law as a compatibility theorem: in the infrared-dominant regime, the metric variation of S_Π and the metric variation of the projected matter functional are related through the local multiplier field, yielding a constraint $\delta S_\Pi = \int \beta(x)^{-1}\,\delta E_\Pi(x)$. This is not a definition but a structural consequence of the Seeley–DeWitt hierarchy and the induced-gravity normalization established in the companion paper.

We then impose a spectral equilibrium principle: admissible geometries extremize S_Π at fixed projected matter content. The resulting Euler–Lagrange equation is Einstein's field equation $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}$, with the factor $8\pi G$ arising purely from the normalization conventions of the geometric and matter sides. The derivation requires no horizon structure, no Rindler wedge, no Raychaudhuri theorem, and no assumption of local Lorentz invariance at the Planck scale. Einstein's equation emerges as a spectral extremum condition within the effective infrared geometry.

Keywords: spectral entropy, heat kernel, local temperature, induced gravity, Einstein equation, spectral equilibrium, Seeley–DeWitt

Contents

1 Introduction

In a companion paper [1], we established that the projective spectral entropy functional

$$S_\Pi[g] \equiv \tfrac{1}{2}\log \det{}' A_g, \tag{1}$$

defined by a minimal Laplace-type operator $A_g = -\nabla_g^2$ on a four-dimensional Riemannian manifold (M, g), admits a well-defined renormalized metric variation. In the infrared-dominant regime where all dimensionless curvature invariants satisfy $R\ell_\chi^2 \ll 1$, the renormalized spectral stress tensor decomposes as

$$\frac{\delta S_\Pi^{\mathrm{ren}}}{\delta g^{\mu\nu}} = c_{\mathrm{EH}}(\mu)\, G_{\mu\nu} + c_\Lambda(\mu)\, g_{\mu\nu} + \beta_{\mathrm{loc}}\, B_{\mu\nu} + T_{\mu\nu}^{\text{non-loc}} + A_{\mu\nu}, \tag{2}$$

with $c_{\mathrm{EH}} = (16\pi G_N)^{-1}$ identified via induced-gravity renormalization; higher-derivative terms are suppressed by powers of ℓ_χ^2.

The goal of the present paper is to develop the local thermodynamic interpretation of this structure. We show that the diagonal heat kernel $K(x, x; t)$ defines a natural local spectral energy density $u(x; t)$, whose curvature expansion produces a local multiplier field $\beta(x)^{-1}$ with a universal part and a geometric correction proportional to the scalar curvature $R(x)$. This field plays the role of a local inverse temperature, defined purely from spectral data, without thermodynamic input.

We then show that the compatibility between the geometric and matter sides of the metric variation is precisely the condition

$$\delta S_\Pi = \int_\Sigma \beta(x)^{-1}\, \delta E_\Pi(x)\, \mathrm{d}^3 x, \tag{3}$$

which we call the *local spectral first law*. This is not a postulate but a structural consequence of the Seeley–DeWitt hierarchy.

Finally, we impose a spectral equilibrium principle: admissible geometries are those for which S_Π is stationary at fixed projected matter content. The Euler–Lagrange equation of this constrained extremum is Einstein's field equation, obtained without any reference to local horizons, Rindler observers, or causal structure.

Relation to Jacobson's derivation. Jacobson [4] derived Einstein's equation by imposing the Clausius relation $\delta S = \delta Q/T$ on all local Rindler horizons, using the Bekenstein–Hawking entropy proportional to area. The present derivation is structurally parallel but differs in three essential ways: (i) the entropy functional is S_Π, not an area functional; (ii) the temperature field $\beta(x)^{-1}$ is defined from spectral data, not from Rindler kinematics; (iii) the field equation arises from a variational extremum, not from a horizon equilibrium condition. The relation to Jacobson's derivation is discussed further in Section 6.

Structure of the paper. Section 2 develops the local spectral energy density and the multiplier field $\beta(x)$. Section 3 establishes the local spectral first law as a compatibility theorem. Section 4 formulates the spectral equilibrium principle and derives the Einstein equation. Section 5 verifies the normalization of the factor $8\pi G$. Section 6 discusses the relation to previous thermodynamic derivations. Section 7 collects limitations and outlook.

2 Spectral locality and the temperature multiplier field

2.1 Local spectral energy density

Let A_g be a positive Laplace-type operator on (M, g). Denote by $K(x, y; t)$ its heat kernel, satisfying

$$(\partial_t + A_g)K(x, y; t) = 0, \qquad K(x, y; 0^+) = \delta_g(x, y). \tag{4}$$

The diagonal $u_0(x; t) \equiv K(x, x; t)$ is a local scalar density encoding the spectral weight at point x and diffusion scale t.

Definition 2.1 (Local spectral energy density)**.** The *local spectral energy density* at point x and scale t is

$$u(x; t) \equiv -\partial_t \log K(x, x; t). \tag{5}$$

This definition is minimal: it depends only on local diagonal heat-kernel data. The sign convention ensures $u(x; t) > 0$ in the short-time regime, consistent with an energy interpretation. The expression is also the natural local analogue of the canonical relation $E = -\partial_\beta \log Z$ in statistical mechanics, once t is identified with an inverse spectral resolution parameter.

2.2 Seeley–DeWitt expansion of $u(x; t)$

For the minimal scalar Laplacian $A_g = -\nabla_g^2$ in dimension $d = 4$, the short-time expansion of the diagonal heat kernel reads [5, 6]

$$K(x, x; t) \sim \frac{1}{(4\pi t)^2}\big(a_0(x) + a_2(x)\, t + a_4(x)\, t^2 + \cdots\big) \quad \text{as } t \to 0^+, \tag{6}$$

with Seeley–DeWitt coefficients

$$a_0(x) = 1, \tag{7}$$

$$a_2(x) = \tfrac{1}{6}R(x), \tag{8}$$

$$a_4(x) = \frac{1}{(4\pi)^2 \cdot 360}\big(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu} + \tfrac{5}{3}R^2\big). \tag{9}$$

Taking the logarithm of (6) and differentiating with respect to t yields:

$$\partial_t \log K(x, x; t) = -\frac{2}{t} + \partial_t \log\big(a_0 + a_2\, t + a_4\, t^2 + \cdots\big). \tag{10}$$

For the minimal Laplacian, $a_0 = 1$ and $a_2 = \frac{1}{6}R$, so

$$\partial_t \log\big(1 + \tfrac{1}{6}R\, t + O(t^2)\big) = \tfrac{1}{6}R + O(t). \tag{11}$$

Therefore, by Definition 2.1:

Proposition 2.2 (Curvature expansion of u)**.** *For the minimal scalar Laplacian in dimension four,*

$$u(x; t) = \frac{2}{t} - \frac{1}{6}R(x) + O(t). \tag{12}$$

The leading term is universal and UV-divergent; the first geometric correction is proportional to the scalar curvature with coefficient fixed by a_2.

2.3 The local multiplier field $\beta(x)$

We now evaluate $u(x;t)$ at a fixed spectral scale $t = t_*$, defined as the admissibility scale introduced in [1, 2] as the infrared threshold below which the spectral description is consistent.

Definition 2.3 (Local spectral multiplier field)**.** The *local spectral multiplier field* $\beta(x)^{-1}$ is defined by

$$\beta(x)^{-1} \equiv u(x;t_*) = \frac{2}{t_*} - \frac{1}{6}R(x) + O(t_*). \tag{13}$$

Remark 2.4 (Sign convention and decomposition)**.** Equation (13) may appear to give a *negative* curvature correction, whereas the literature on local temperature often writes a *positive* geometric correction. The sign is consistent and arises from the definition $u = -\partial_t \log K$ together with the expansion $\log K \sim -2\log t + \frac{1}{6}Rt + \cdots$, which gives $\partial_t \log K \sim -2/t + \frac{1}{6}R$, and hence $u \sim 2/t - \frac{1}{6}R$.

When we write the decomposition with $\beta_0^{-1} \equiv 2/t_*$, we have

$$\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(t_*). \tag{14}$$

On a positively curved manifold ($R > 0$), the local multiplier is *smaller* than the flat reference β_0^{-1}: curvature reduces the effective spectral temperature relative to flat space. This sign is physically consistent with the Seeley–DeWitt interpretation: positive curvature lowers the local spectral density, hence lowers the effective temperature.

Remark 2.5 (Status of t_* and β_0^{-1})**.** The scale t_* is a spectral admissibility parameter, not a physical temperature. Its value is fixed by the spectral resolution of the relational system, as discussed in [2]. The universal part $\beta_0^{-1} = 2/t_*$ is a UV quantity reflecting the spectral density of states at the cutoff. It is scheme-dependent and carries no intrinsic thermodynamic meaning at this stage. Only the geometric correction $\frac{1}{6}R(x)$ is covariant and physically significant: it encodes local curvature as a deviation from the flat-space multiplier.

Remark 2.6 (Decomposition structure)**.** The decomposition (14) has the form

$$\beta(x)^{-1} = \underbrace{\beta_0^{-1}}_{\text{universal (UV)}} - \underbrace{\tfrac{1}{6}R(x)}_{\text{geometric correction}}.$$

The geometric correction carries a negative sign: positive curvature reduces the local spectral multiplier relative to the flat-space reference β_0^{-1}. The correction is entirely controlled by the a_2 Seeley–DeWitt coefficient, which is precisely the coefficient that generates the Einstein–Hilbert counterterm via pole subtraction in the companion paper. The two papers are therefore connected at the level of the same spectral coefficient.

3 Local spectral first law as a compatibility theorem

3.1 Projected matter energy

We define the projected matter energy on a spatial slice Σ independently of the spectral multiplier field. Let $W_\Pi[g,\psi]$ be the effective matter action associated with the admissible projected sector. Its metric variation defines the projected stress tensor $T^{(\Pi)}_{\mu\nu}$ via the standard covariant identity

$$\delta W_\Pi[g,\psi] = \tfrac{1}{2}\int_M \mathrm{d}^4x\,\sqrt{g}\,T^{(\Pi)}_{\mu\nu}\,\delta g^{\mu\nu}. \tag{15}$$

On a chosen spatial slice Σ with future-pointing unit normal n^μ and local Killing vector ξ^μ, the projected matter energy is defined by

$$E_\Pi[\Sigma] \equiv \int_\Sigma T^{(\Pi)}_{\mu\nu}\,\xi^\mu n^\nu\,\mathrm{d}\Sigma, \tag{16}$$

and its metric variation is

$$\delta E_\Pi = \tfrac{1}{2}\int_\Sigma T^{(\Pi)}_{\mu\nu}\,\delta g^{\mu\nu}\,\mathrm{d}\Sigma. \tag{17}$$

This definition is independent of $\beta(x)$ and of the heat-kernel structure.

3.2 Local spectral first law

Proposition 3.1 (Local spectral first law — compatibility condition)**.** *In the infrared regime $R\ell_\chi^2 \ll 1$, and at leading order in the Seeley–DeWitt derivative expansion, the metric variation of the renormalized spectral entropy satisfies*

$$\delta S^{\mathrm{ren}}_\Pi = \int_\Sigma \beta(x)^{-1}\,\delta E_\Pi(x)\,\mathrm{d}^3x \tag{18}$$

as a structural compatibility identity between the geometric and matter sides of the variational problem. This identity holds if and only if the geometry satisfies the Einstein equation. It is therefore an equivalence condition, not an independent postulate.

Proof (compatibility argument). From [1], the infrared-dominant local part of $\delta S^{\mathrm{ren}}_\Pi$ contains the Einstein tensor:

$$\delta S^{\mathrm{ren}}_\Pi \supset \frac{1}{16\pi G_N}\int_M \mathrm{d}^4x\,\sqrt{g}\,G_{\mu\nu}\,\delta g^{\mu\nu} + O(R^2\ell_\chi^2). \tag{19}$$

The Bianchi identity $\nabla^\mu G_{\mu\nu} = 0$ implies that the integrand $G_{\mu\nu}\delta g^{\mu\nu}$ admits a local density representation on Σ once $\delta g^{\mu\nu}$ is supported in a compact region. On the matter side, from (17):

$$\delta E_\Pi = \tfrac{1}{2}\int_\Sigma T^{(\Pi)}_{\mu\nu}\,\delta g^{\mu\nu}\,\mathrm{d}\Sigma. \tag{20}$$

Requiring (18) to hold with $\beta(x)^{-1}$ as in (14), then matching the two sides at leading IR order requires

$$\frac{1}{16\pi G_N}\,G_{\mu\nu} = \beta(x)^{-1}\cdot\tfrac{1}{2}\,T^{(\Pi)}_{\mu\nu}, \tag{21}$$

which is consistent with the Einstein equation when $\beta(x)^{-1}$ is evaluated at the equilibrium value. In spectral units where $\beta_0^{-1} = 1$, the curvature correction $-\tfrac{1}{6}R$ is of order $R\ell_\chi^2$ and is absorbed into the subleading remainder (see Remark 5.2). The identity (18) is therefore a structural compatibility condition: it holds if and only if the geometry satisfies the Einstein equation at leading infrared order. This establishes (18) as a compatibility condition rather than a postulate. □

Remark 3.2 (Nature of the identity)**.** Equation (18) is not a definition of $\beta(x)$ from δS_Π and δE_Π: both sides are independently defined. $\beta(x)^{-1}$ is defined from the heat kernel via Definition 2.3; δE_Π is defined from the matter action via (17). The identity (18) is therefore a non-trivial constraint, not a tautology.

4 Spectral equilibrium principle and Einstein equation

4.1 Formulation of the equilibrium principle

We now impose the following variational principle.

Definition 4.1 (Spectral equilibrium)**.** An admissible geometry g is in *spectral equilibrium* if it is a stationary point of the functional

$$\mathcal{F}[g, \psi] \equiv S_\Pi[g] - \beta_*^{-1} W_\Pi[g, \psi] \tag{22}$$

under arbitrary compactly supported metric variations $\delta g^{\mu\nu}$, with β_*^{-1} a constant Lagrange multiplier.

The functional $\mathcal{F}$ is the spectral analogue of a free energy: it extremizes entropic gain against the projected matter cost. The Lagrange multiplier β_*^{-1} enforces the constraint that the projected matter content W_Π is held fixed along admissible deformations.

Remark 4.2 (Nature of β_*^{-1})**.** The global multiplier β_*^{-1} in (22) is distinct from the local field $\beta(x)^{-1}$ defined in Section 2. The local field encodes spatial curvature variation; the global multiplier encodes the overall scale of the matter-geometry coupling. Their relation is discussed in Section 5.

4.2 Euler–Lagrange equation

Theorem 4.3 (Einstein equation as spectral equilibrium)**.** *In the infrared-dominant regime $R\ell_\chi^2 \ll 1$, the Euler–Lagrange equation of the spectral equilibrium condition $\delta\mathcal{F} = 0$ is*

$$G_{\mu\nu} = 8\pi G_N \, T_{\mu\nu}^{(\Pi)}, \tag{23}$$

up to a cosmological term and corrections suppressed by $O(R\ell_\chi^2)$.

Proof. Varying $\mathcal{F}[g, \psi]$ with respect to $g^{\mu\nu}$:

$$\delta\mathcal{F} = \delta S_\Pi - \beta_*^{-1} \, \delta W_\Pi = 0. \tag{24}$$

Substituting (19) and (15):

$$\frac{1}{16\pi G_N} \int_M \sqrt{g} \, G_{\mu\nu} \, \delta g^{\mu\nu} = \frac{\beta_*^{-1}}{2} \int_M \sqrt{g} \, T_{\mu\nu}^{(\Pi)} \, \delta g^{\mu\nu} + O(R\ell_\chi^2). \tag{25}$$

Since this holds for arbitrary compactly supported $\delta g^{\mu\nu}$, the fundamental lemma of the calculus of variations gives pointwise:

$$\frac{1}{16\pi G_N} G_{\mu\nu} = \frac{\beta_*^{-1}}{2} T_{\mu\nu}^{(\Pi)} + O(R\ell_\chi^2). \tag{26}$$

For $\beta_*^{-1} = 1$ (see Section 5), this gives

$$G_{\mu\nu} = 8\pi G_N \, T_{\mu\nu}^{(\Pi)} + O(R\ell_\chi^2). \tag{27}$$

□

Remark 4.4 (Absence of causal assumptions)**.** The derivation requires no horizon structure, no local Rindler observers, no Raychaudhuri equation, and no assumption on causal structure. The field equation emerges from a purely variational condition on the spectral functional $\mathcal{F}$, defined on Riemannian manifolds. Its extension to Lorentzian signature is a separate question, discussed in Section 7.

5 Normalization of the factor $8\pi G$

5.1 Two independent normalizations

The factor $8\pi G$ in equation (23) is entirely determined by two independent normalization conventions:

Geometric normalization. The companion paper [1] establishes that the infrared-dominant local part of the renormalized spectral entropy variation contains the Einstein tensor with coefficient $1/(16\pi G_N)$:

$$\left.\frac{\delta S_\Pi^{\rm ren}}{\delta g^{\mu\nu}}\right|_{\rm IR,\,local} = \frac{1}{16\pi G_N}\,G_{\mu\nu}. \tag{28}$$

This coefficient arises from the a_2 pole subtraction in the zeta-regularized determinant and the induced-gravity identification $G_N \sim 16\pi^2\ell_\chi^2$.

Matter normalization. The projected stress tensor is defined via the standard variational identity

$$T_{\mu\nu}^{(\Pi)} = \frac{2}{\sqrt{g}}\frac{\delta W_\Pi}{\delta g^{\mu\nu}}, \tag{29}$$

which carries the universal factor 2 by convention. Equivalently, $\delta W_\Pi = \frac{1}{2}\int \sqrt{g}\,T_{\mu\nu}^{(\Pi)}\delta g^{\mu\nu}$.

5.2 The factor $8\pi G$ as a product of normalizations

Proposition 5.1 (Normalization identity)**.** *The equilibrium condition $\delta\mathcal{F} = 0$ with $\beta_*^{-1} = 1$ gives*

$$\frac{1}{16\pi G_N}\,G_{\mu\nu} = \tfrac{1}{2}\,T_{\mu\nu}^{(\Pi)}, \tag{30}$$

which is equivalent to $G_{\mu\nu} = 8\pi G_N\,T_{\mu\nu}^{(\Pi)}$. The numerical factor $8\pi G_N$ arises as the unique product of two independent normalizations:

$$8\pi G_N = \frac{1}{(16\pi G_N)^{-1}\times\frac{1}{2}} = \frac{1}{(16\pi G_N)^{-1}}\cdot 2\cdot 1, \tag{31}$$

where the factor $(16\pi G_N)^{-1}$ comes from the geometric normalization (28) and the factor $\frac{1}{2}$ from the stress-tensor definition (29). More transparently: from $\frac{1}{16\pi G_N}G_{\mu\nu} = \frac{1}{2}T_{\mu\nu}$ one reads off directly $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$. No other factor enters.

5.3 Status of $\beta_*^{-1} = 1$

Setting $\beta_*^{-1} = 1$ in the constrained functional (22) is not a dynamical assumption. It is a normalization choice equivalent to measuring energy in units where the spectral entropy and the projected matter functional are coupled with unit strength. Concretely, it amounts to absorbing β_*^{-1} into the definition of $T_{\mu\nu}^{(\Pi)}$ as the effective stress tensor of the admissible projected sector, so that W_Π is defined with the appropriate normalization.

Alternatively, one may retain β_*^{-1} as a free parameter and interpret $\beta_*^{-1} \neq 1$ as a departure from canonical spectral equilibrium. In that case, (26) generalizes to $G_{\mu\nu} = 8\pi G_N\,\beta_*^{-1}\,T_{\mu\nu}^{(\Pi)}$, which describes an off-equilibrium spectral geometry with an effective renormalization of the matter coupling.

5.4 Relation between β_*^{-1} and the local field $\beta(x)^{-1}$

The local field $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x)$ and the global Lagrange multiplier β_*^{-1} play distinct roles. The local field encodes spatially varying curvature corrections to the spectral multiplier; the global multiplier controls the overall strength of the entropy-matter coupling in the equilibrium functional.

Their connection arises when the equilibrium condition is imposed locally rather than globally. Requiring $\delta\mathcal{F}$ to vanish not only for global metric variations but also for local ones supported near a point x replaces β_*^{-1} by $\beta(x)^{-1}$ in (26), yielding

$$G_{\mu\nu}(x) = 8\pi G_N \, \beta(x)^{-1} \, T^{(\Pi)}_{\mu\nu}(x) + O(R\ell_\chi^2). \tag{32}$$

To display the structure of the curvature correction transparently, we choose *spectral units* in which $\beta_0^{-1} = 1$.

Remark 5.2 (Gauge choice $\beta_0^{-1} = 1$ and the spectral scale t_*)**.** Setting $\beta_0^{-1} = 1$ corresponds, via $\beta_0^{-1} = 2/t_*$, to the choice $t_* = 2$ in spectral units. This is a *gauge choice for the admissibility scale*: it fixes the unit of spectral resolution but does not affect any physical observable. In particular, the induced Newton coupling $G_N \sim 16\pi^2\ell_\chi^2$ established in [1] is independent of t_*, since it arises from the a_2 pole residue, which is a UV quantity fixed by the renormalization scheme.

In these units, the local field simplifies to

$$\beta(x)^{-1} = 1 - \frac{1}{6}R(x) + O(\nabla^2 R), \tag{33}$$

and the locally imposed equilibrium condition (32) becomes

$$G_{\mu\nu}(x) = 8\pi G_N \big(1 - \tfrac{1}{6}R(x)\big)\, T^{(\Pi)}_{\mu\nu}(x) + O(R\ell_\chi^2). \tag{34}$$

At zeroth order in $R\ell_\chi^2$, this reduces to the standard Einstein equation. The correction $-\frac{1}{6}R(x)$ is of order $R\ell_\chi^2$ and is absorbed into the $O(R\ell_\chi^2)$ remainder in the weak-curvature regime. Changing the gauge (i.e. changing t_*) rescales β_0^{-1} and the correction simultaneously, leaving the physical content of (32) unchanged.

6 Relation to previous thermodynamic derivations

6.1 Jacobson's derivation

Jacobson [4] derived Einstein's equation by applying the Clausius relation $\delta S = \delta Q/T$ to all local Rindler horizons. The key inputs are: (i) entropy proportional to horizon area, $S = A/(4G)$; (ii) temperature given by the Unruh formula, $T = a/(2\pi)$; (iii) heat flow $\delta Q = \int T_{\mu\nu}\chi^\mu \mathrm{d}\Sigma^\nu$ across the horizon; (iv) the Raychaudhuri equation linking area change to matter flux.

The present derivation parallels Jacobson's logical structure — entropy, temperature, energy, equilibrium condition — but replaces each ingredient with a spectral counterpart: (i) entropy is $S_\Pi = \frac{1}{2}\log\det' A_g$ (not an area functional); (ii) temperature is the local spectral multiplier $\beta(x)^{-1}$ (not Rindler); (iii) energy is the projected matter content E_Π (not a horizon heat flux); (iv) equilibrium is a variational extremum (not a horizon balance condition).

As a consequence, the present derivation is independent of the causal structure of spacetime: it holds on Riemannian manifolds and does not require a Lorentzian horizon. The extension to Lorentzian signature is discussed in Section 7.

6.2 Analogy with local Unruh temperature

In the Unruh effect, the local temperature experienced by a Rindler observer with proper acceleration a is $T_{\mathrm{Unruh}} = a/(2\pi)$. This is kinematic: it depends on the observer's trajectory.

The local multiplier field $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x)$ has a formally analogous structure: a universal part β_0^{-1} plus a local correction. However, the analogy is purely formal. The correction $-\frac{1}{6}R(x)$ is geometric (curvature-driven), not kinematic (acceleration-driven). No horizon is present or implied. The analogy should not be taken to mean that $\beta(x)^{-1}$ is a physical temperature in the Unruh sense: it is a spectral multiplier defined from heat-kernel data, not from a quantum state on a wedge.

6.3 Wald's Noether charge entropy

Wald [10] showed that black hole entropy for higher-derivative theories of gravity is given by a Noether charge, not by the area. The present framework is consistent with Wald's result in the following sense: the spectral entropy S_Π is defined from the full operator A_g, which encodes higher-derivative geometric information through the a_4 and higher Seeley–DeWitt coefficients. In the UV regime (or for $\xi \neq 0$), the dominant contribution to S_Π shifts from the a_2-derived Einstein term to the a_4-derived quadratic curvature terms, whose variation would give a higher-derivative stress tensor consistent with Wald's formula. A detailed comparison lies beyond the scope of the present paper.

7 Discussion and outlook

7.1 Summary

We have shown that the projective spectral entropy functional $S_\Pi[g]$ admits a natural local thermodynamic structure. The diagonal heat kernel defines a local spectral energy density $u(x;t)$ whose Seeley–DeWitt expansion yields a local multiplier field $\beta(x)^{-1} = \beta_0^{-1} - \frac{1}{6}R(x) + O(\nabla^2 R)$ decomposed into a universal UV part and a geometric curvature correction.

The local spectral first law $\delta S_\Pi = \int \beta^{-1}\delta E_\Pi$ is established as a compatibility theorem, not a definition: both sides are independently defined from spectral and matter data respectively.

The spectral equilibrium principle — extremizing S_Π at fixed projected matter content — yields Einstein's equation $G_{\mu\nu} = 8\pi G_N T_{\mu\nu}$ as an Euler–Lagrange condition. The factor $8\pi G_N$ is fixed by the product of two universal normalizations from the geometric and matter sides, with no free parameter.

7.2 Limitations

Euclidean setting. The entire construction is performed on Riemannian manifolds. The extension to Lorentzian signature requires analytic continuation of the heat kernel, control over the retarded Green function, and a definition of $u(x;t)$ in the hyperbolic setting. This is the content of the Lorentzian completion program described in [1].

Static foliation. The projected matter energy $E_\Pi[\Sigma]$ is defined on a fixed spatial slice. A fully covariant formulation would require a foliation-independent definition, likely via the covariant phase space formalism.

Higher Seeley–DeWitt corrections. The equilibrium condition (23) holds at leading IR order. In the regime $R\ell_\chi^2 \sim 1$, higher Seeley–DeWitt contributions (from a_4 and beyond) become comparable, and the field equation receives quadratic curvature corrections of the form $G_{\mu\nu} + \ell_\chi^2 B_{\mu\nu} + \cdots = 8\pi G_N T_{\mu\nu}$.

Matter coupling. The projected stress tensor $T_{\mu\nu}^{(\Pi)}$ is defined formally from W_Π. An explicit construction of W_Π in terms of standard matter fields requires identifying the admissible projected sector for each matter species. This is addressed in the Cosmochrony framework [3] but not derived here.

7.3 Outlook

The present framework opens several directions. First, the Lorentzian completion of Paper 3 will test whether the equilibrium condition survives analytic continuation and whether it reproduces linearized gravitational wave propagation. Second, the higher Seeley–DeWitt regime (a_4 dominant, $\xi = 1/6$) may provide a spectral derivation of higher-curvature gravity theories. Third, the connection between the local field $\beta(x)^{-1}$ and the local Unruh effect could be made precise if a Lorentzian horizon is introduced as a special case of the general framework.

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